THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 25%	Change/original × 100 = 20/80×100
2	**(c)** 25%	18/72×100
3	**(b)** 20%	(SP−CP)/CP×100 = 300/1500×100
4	**(d)** 20%	Discount = 500; 500/2500×100
5	**(a)** 2100	GST = 100; bill = 2100
6	**(d)** 1050	Amount = 5000(1+10/100)² = 6050; CI = A − P
7	**(a)** 20%	Change/original × 100 = 50/250×100
8	**(d)** 25%	35/140×100
9	**(b)** 10%	(SP−CP)/CP×100 = 240/2400×100
10	**(c)** 10%	Discount = 180; 180/1800×100

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$36/90 \times 100$ (1) $= 40\%$ (1).	1+1
12	Loss = 15% of 24000 = ₹ 3600 (1); SP = ₹ 20400 (1).	1+1
13	SI = ₹ 720 (1); R = SI×100/(P×T) = 6% p.a. (1).	1+1
14	$27/120 \times 100$ (1) $= 22.5\%$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	GST = 18% of 45000 = ₹ 8100 (2); bill = 45000 + 8100 = ₹ 53100 (1).	2+1
16	A = P(1+R/100)² = 12500×108/100×108/100 = ₹ 14580 (2); CI = ₹ 2080 (1).	2+1
17	Change = 4000 (1); % = 4000/40000×100 (1) = 10% (1).	1+1+1
	OR: GST = 12% of 36000 = ₹ 4320 (2); bill = 36000 + 4320 = ₹ 40320 (1).	2+1
18	A = P(1+R/100)² = 16000×125/100×125/100 = ₹ 25000 (2); CI = ₹ 9000 (1).	2+1
	OR: Change = 2500 (1); % = 2500/25000×100 (1) = 10% (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A = P(1+R/100)³ = ₹ 10648 (2); CI = ₹ 2648 (1); SI = 8000×10×3/100 = ₹ 2400 (1); difference = ₹ 248 (1).	2+1+1+1
	OR: Discount = ₹ 3000 (1); discounted price = ₹ 27000 (1); GST = 18% of it = ₹ 4860 (1); payable = ₹ 31860 (2).	1+1+1+2
20	A = P(1+R/100)³ = ₹ 8640 (2); CI = ₹ 3640 (1); SI = 5000×20×3/100 = ₹ 3000 (1); difference = ₹ 640 (1).	2+1+1+1
	OR: Discount = ₹ 5000 (1); discounted price = ₹ 20000 (1); GST = 12% of it = ₹ 2400 (1); payable = ₹ 22400 (2).	1+1+1+2

Mistakes to avoid:

- Calculating profit or loss percent on the selling price; it is always on the **cost price**.
- Calculating discount on the selling price; it is on the **marked price**.
- Adding two successive percent changes: $+10\%$ then -10% is a 1% decrease, not 0%.
- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1680	GST = 180; bill = 1680
2	**(b)** 880	Amount = 2000(1+20/100)² = 2880; CI = A − P
3	**(d)** 25%	Change/original × 100 = 15/60×100
4	**(b)** 15%	9/60×100
5	**(d)** 10%	(SP−CP)/CP×100 = 320/3200×100
6	**(c)** 15%	Discount = 144; 144/960×100
7	**(c)** 944	GST = 144; bill = 944
8	**(b)** 1025	Amount = 10000(1+5/100)² = 11025; CI = A − P
9	**(c)** 15%	Change/original × 100 = 60/400×100
10	**(b)** 60%	48/80×100

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Loss = 8% of 18500 = ₹ 1480 (1); SP = ₹ 17020 (1).	1+1
12	SI = ₹ 1200 (1); R = SI×100/(P×T) = 8% p.a. (1).	1+1
13	$45/75 \times 100$ (1) $= 60\%$ (1).	1+1
14	Loss = 12% of 32000 = ₹ 3840 (1); SP = ₹ 28160 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	GST = 18% of 62000 = ₹ 11160 (2); bill = 62000 + 11160 = ₹ 73160 (1).	2+1
16	A = P(1+R/100)² = 4000×115/100×115/100 = ₹ 5290 (2); CI = ₹ 1290 (1).	2+1
17	Change = 5600 (1); % = 5600/56000×100 (1) = 10% (1).	1+1+1
	OR: GST = 5% of 28500 = ₹ 1425 (2); bill = 28500 + 1425 = ₹ 29925 (1).	2+1
18	A = P(1+R/100)² = 30000×110/100×110/100 = ₹ 36300 (2); CI = ₹ 6300 (1).	2+1
	OR: Change = 4200 (1); % = 4200/84000×100 (1) = 5% (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A = P(1+R/100)³ = ₹ 19965 (2); CI = ₹ 4965 (1); SI = 15000×10×3/100 = ₹ 4500 (1); difference = ₹ 465 (1).	2+1+1+1
	OR: Discount = ₹ 6000 (1); discounted price = ₹ 34000 (1); GST = 18% of it = ₹ 6120 (1); payable = ₹ 40120 (2).	1+1+1+2
20	A = P(1+R/100)³ = ₹ 21600 (2); CI = ₹ 9100 (1); SI = 12500×20×3/100 = ₹ 7500 (1); difference = ₹ 1600 (1).	2+1+1+1
	OR: Discount = ₹ 900 (1); discounted price = ₹ 17100 (1); GST = 28% of it = ₹ 4788 (1); payable = ₹ 21888 (2).	1+1+1+2

Mistakes to avoid:

- Calculating profit or loss percent on the selling price; it is always on the **cost price**.
- Calculating discount on the selling price; it is on the **marked price**.
- Adding two successive percent changes: $+10\%$ then -10% is a 1% decrease, not 0%.
- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Fractions in Disguise** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 20%	(SP−CP)/CP×100 = 250/1250×100
2	**(a)** 15%	Discount = 600; 600/4000×100
3	**(d)** 4480	GST = 980; bill = 4480
4	**(a)** 2580	Amount = 8000(1+15/100)² = 10580; CI = A − P
5	**(c)** 20%	Change/original × 100 = 30/150×100
6	**(b)** 37.5%	21/56×100
7	**(b)** 20%	(SP−CP)/CP×100 = 170/850×100
8	**(d)** 20%	Discount = 150; 150/750×100
9	**(b)** 1260	GST = 60; bill = 1260
10	**(b)** 3600	Amount = 6400(1+25/100)² = 10000; CI = A − P

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	SI = ₹ 2400 (1); R = SI×100/(P×T) = 5% p.a. (1).	1+1
12	$63/140 \times 100$ (1) $= 45\%$ (1).	1+1
13	Loss = 25% of 9600 = ₹ 2400 (1); SP = ₹ 7200 (1).	1+1
14	SI = ₹ 1260 (1); R = SI×100/(P×T) = 7% p.a. (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	GST = 28% of 52000 = ₹ 14560 (2); bill = 52000 + 14560 = ₹ 66560 (1).	2+1
16	A = P(1+R/100)² = 22500×120/100×120/100 = ₹ 32400 (2); CI = ₹ 9900 (1).	2+1
17	Change = 4800 (1); % = 4800/32000×100 (1) = 15% (1).	1+1+1
	OR: GST = 12% of 15000 = ₹ 1800 (2); bill = 15000 + 1800 = ₹ 16800 (1).	2+1
18	A = P(1+R/100)² = 7200×105/100×105/100 = ₹ 7938 (2); CI = ₹ 738 (1).	2+1
	OR: Change = 7200 (1); % = 7200/72000×100 (1) = 10% (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	A = P(1+R/100)³ = ₹ 26620 (2); CI = ₹ 6620 (1); SI = 20000×10×3/100 = ₹ 6000 (1); difference = ₹ 620 (1).	2+1+1+1
	OR: Discount = ₹ 6000 (1); discounted price = ₹ 44000 (1); GST = 5% of it = ₹ 2200 (1); payable = ₹ 46200 (2).	1+1+1+2
20	A = P(1+R/100)³ = ₹ 10800 (2); CI = ₹ 4550 (1); SI = 6250×20×3/100 = ₹ 3750 (1); difference = ₹ 800 (1).	2+1+1+1
	OR: Discount = ₹ 6000 (1); discounted price = ₹ 18000 (1); GST = 12% of it = ₹ 2160 (1); payable = ₹ 20160 (2).	1+1+1+2

Mistakes to avoid:

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- Adding two successive percent changes: $+10\%$ then -10% is a 1% decrease, not 0%.
- Using the simple interest formula for compound interest, or forgetting to subtract P from the amount to get CI.
- Applying GST on the marked price when the question says it is charged after the discount.
- Writing a percentage without dividing by 100 in calculations (e.g. 12% of 2000 as 12×2000).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.